B.Sc. (Computer Science)
CBCS Pattern with Effect from the Academic Year 2022-2023

SEMESTER – III
Operating Systems with Linux

UNIT I

Operating System: Concept, Components of Operating System, Operating System Operations, Protection and Security. Computing Environment. Abstract View of OS: User view, System View, Operating System Services, System Calls: Concept, Types of System Calls. Computer System Architecture: Single-Processor Systems, Multiprocessor Systems. Types of Operating Systems: Batch Operating System, Multi-Programmed Operating System, Time-Shared Operating System, Real Time Operating System, Distributed Operating Systems. Process Management: Process Concept, Operation on Processes, Cooperating Processes, Inter-Process Communication, Threads. Linux Operating System: Introduction to Linux OS, Basic Commands of Linux OS.

UNIT II

Process Synchronization: Introduction, The Critical-Section Problem with solution, Bakery Algorithm, Synchronization hardware, Semaphores, Semaphores Implementation, Classical Problems of Synchronization with algorithms, Critical Regions, Monitors. CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling algorithms, Multilevel Queue Scheduling, Multilevel Feedback Queue Scheduling. Linux Operating System: Process Management Commands and System Calls.

UNIT III

Deadlock: System Models, Deadlock Characterization, Resource Allocation Graph, Deadlock Prevention, Avoidance, Detection and Recovery, Banker's algorithm. Memory Management: Main Memory: Contiguous Memory Allocation, Fragmentation, Paging, And Segmentation. Virtual Memory: Demand Paging, Page Replacement, Page replacement algorithm, Allocation of frames, Thrashing. Linux Operating System: Memory Management Commands and System Calls.

UNIT IV

File, Devices and Secondary Storage Management: File-System Interface: Concepts, Access Methods, Directory and Disk Structure. File-System Structure, File-System Implementation, Directory Implementation, Allocation Methods, Free-Space Management. Devices: Types of devices, Channels and Control Unit, Multiple Paths, Block Multiplexing. Secondary Storage: Mass-Storage Structure, Disk Structure, Disk Scheduling Algorithms, Disk Management, RAID structure of disk. Linux Operating System: File Management Commands and System Calls.

TEXT BOOKS:

1. Silberschatz, Galvin, Greg, "Operating System Concepts", Wiley and Sons, 9th Edition, 2015.
2. Sumitabha Das, "Unix concept and Programming", McGraw Hill education, 4th Edition, 2015.

REFERENCE BOOKS:

1. Godbole, Achyut, "Operating System", McGraw-Hill Education, 2nd Edition, 2005.
2. William Stallings, "Operating System: Internals and Design Principles", Person, 9th Edition, 2018.
3. A. S. Tanenbaum, "Modern Operating Systems ", Pearson, 3rd Edition, 2007.
4. Kenneth H. Rosenet al, "UNIX: The Complete Reference", McGraw-Hill/Osborne, 6th Edition, 2017.